DRAFT SCHEME

Total Pages: 4

Scheme of Valuation/Answer Key

Scheme of evaluation (marks in brackets) and answers of problems/key

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

THIRD SEMESTER B.TECH DEGREE EXAMINATION, DECEMBER 2022

Course Code: CST201

Course Name: DATA STRUCTURES

Max. Marks: 100 Duration: 3 Hours

PART A

	Answer all questions. Each question carries 3 marks	Marks
1	Algorithm 1.5 Marks, Frequency counting Method 1.5 Marks	(3)
2	Space complexity- Instruction Space, Environmental Stack, Data Space	(3)
3	Structure definition 1 Marks, Figure showing the representation 2 Marks	(3)
4	Postfix expression A B C * + D E / H * - Correct output 1 Mark, Table showing the conversion steps 2 Marks	(3)
5	Definition 1 mark, example 1 Mark, Use 1 mark	(3)

6	Algorithm Count(Head)
	{
	Temp = Head, count = 0
	While (temp !Null)
	count =count + 1
	Temp = Temp->next
	Wend
	Print count
		(3)

7	BST Explanation 1 Marks, properties 1 mark, Example (figure) 1 Mark	(3)
8	Adjacency Matrix 1.5 Marks	(3)

v	V1	v2	v3	v4	v5	v6	v7
v1	0	1	1	0	0	0	0
V2	1	0	0	1	1	0	0
V3	1	0	0	1	0	1	0

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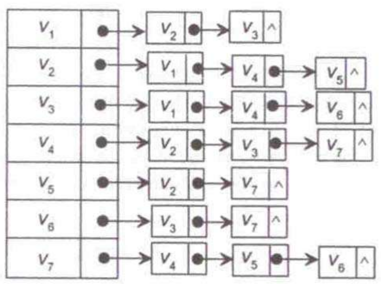
V4	0	1	1	0	0	0	7
V5	0	1	0	0	0	0	1
V6	0	0	1	0	0	0	1
V7	0	0	0	1	1	1	0

Adjacency List 1.5 Marks



	9	Insertion Sort Algorithm 3 Marks
		Algorithm InsertioSort(A[ ], n)
		{
		1. Repeat for i=1 to n-1
		2. Set key= a[i]
		3. j=i-1
		4. while j>=0 & a[j]>key Repeat step 5 and 6
		5. a[j+1]=a[j]
		6. j=j-1
		7. End while
		8. a[j+1]=key
		9. End repeat
		}
			(3)

	10	Any two commonly used hash functions(2 * 1.5 Marks)	(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

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Module 1			
11	a	Big-Oh Notation, Big-Omega Notation, Big-Theta Notation. Definition, Graph, Example (3 * 3 Marks)	(9)

	b	for (i = 0; i< n; i++) --------------------------- (n+1)
		for (j = 0; j< n; j++) ---------------------(n+1)(n+1)
		x = x + 1;. …………………..(n+1)(n+1)
		………………………
		2n^2 +5n+3
			(5)

12	a	Requirements Design Analysis Refinement and Coding Verification (8 Marks)	(8)
	b	Time complexity (2 Marks), output O(n^2) (1 Mark), Derivation 3 Marks	(6)
Module 2			
13	a	Algorithm 5 Marks, Demonstration 4 Marks	(8)
	b	Stack – 3.5 Marks. Status of stack 2.5 Marks	(6)
14	a	Algorithm 4 Marks. Evaluation steps – 3 Marks, Output (-15) 1 Mark.	(8)
	b	Circular Queue with figure 2 Marks. Insertion 2 Marks, Deletion 2 Marks	(6)
Module 3			
15	a	Advantages of linked list over arrays (2 Marks)? Insertion using linked list (3 Marks) Deletion using linked list (3 Marks)	(8)
	b	Status of memory allocation first-fit (2 marks) , best-fit (2 marks), and worst-fit(2 marks)	(6)
16	a	Polynomial addition algorithm using linked list (6 Marks) Example (4 Marks)	(10)
	b	Comparison (4 Marks)	(4)
Module 4			
17	a	BFS Algorithm (5 Marks) BFS illustration (3 Marks)	(8)

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	b	Root node at index 1, If a node at index i then its left child is at 2i and is right child is at 2i + 1 (Array representation) 2 Marks. Tree construction 3 Marks	(6)
18	a	BST 2 Marks, Algorithm 5 Marks, Example 3 Marks	(10)
	b	Comparison between binary tree and full binary tree – 4 Marks	(4)
		Module 5	
19	a	Merge Sort algorithm (6 Marks)Demonstration (4 Marks)	(10)
	b	Definition 2 Marks, Example 2 Marks	(4)
20	a	Closed hashing and its variations(5 Marks), Open Chaining (3 Marks)	(8)
	b	Hash table – 6 Marks	(6)